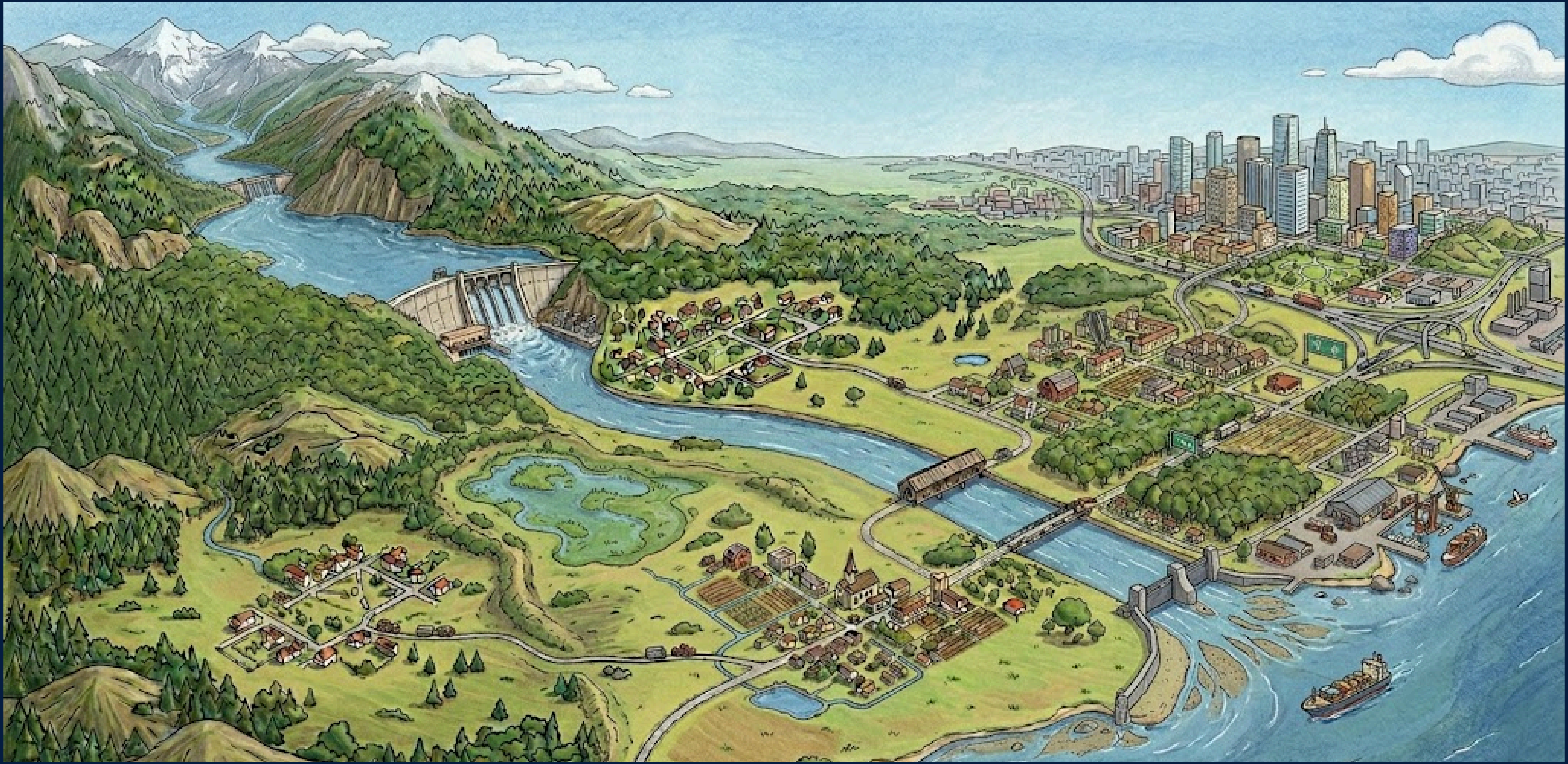
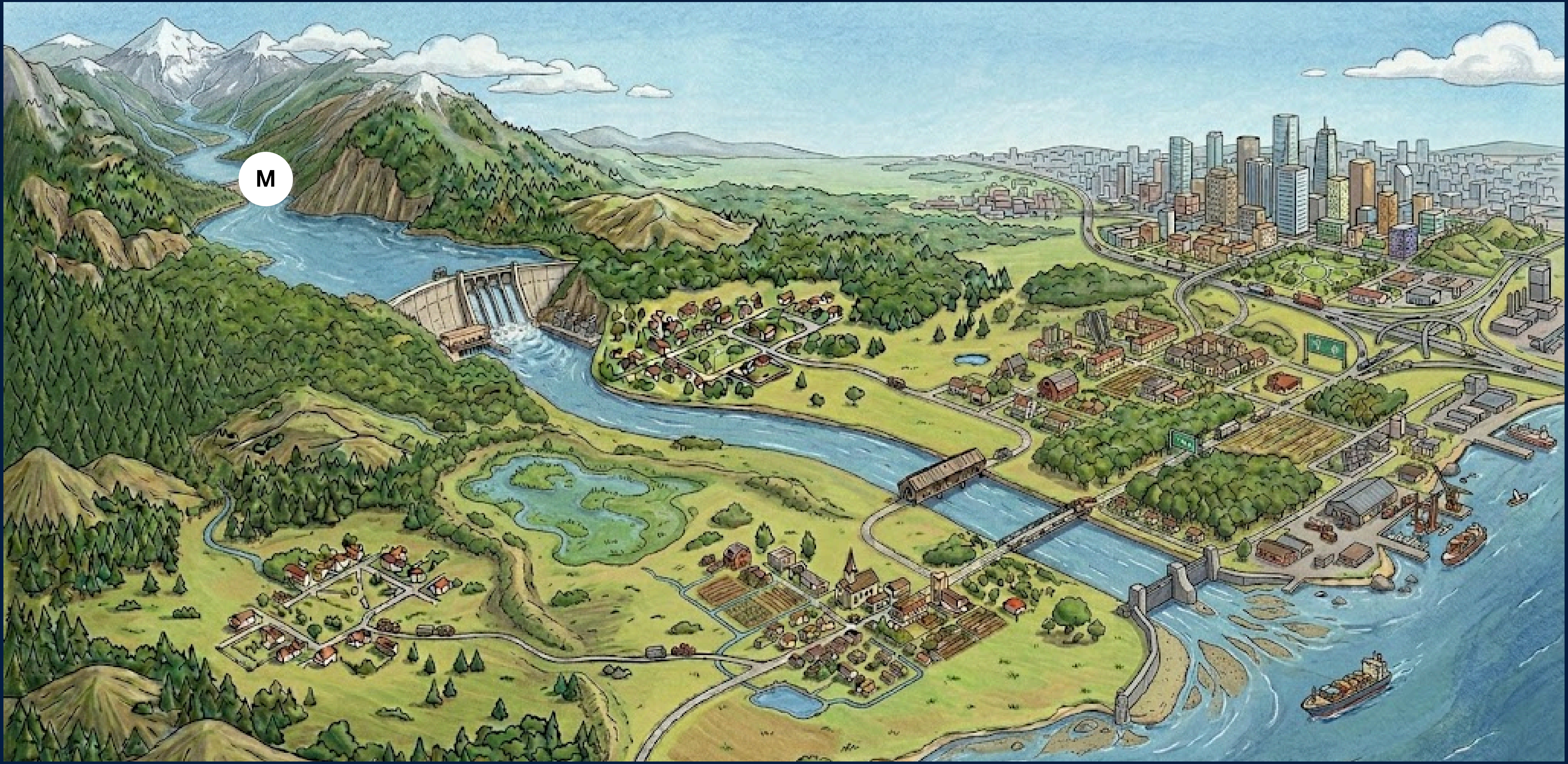
RE-MESH

IoT-Based Flood Detection & Disaster Management System

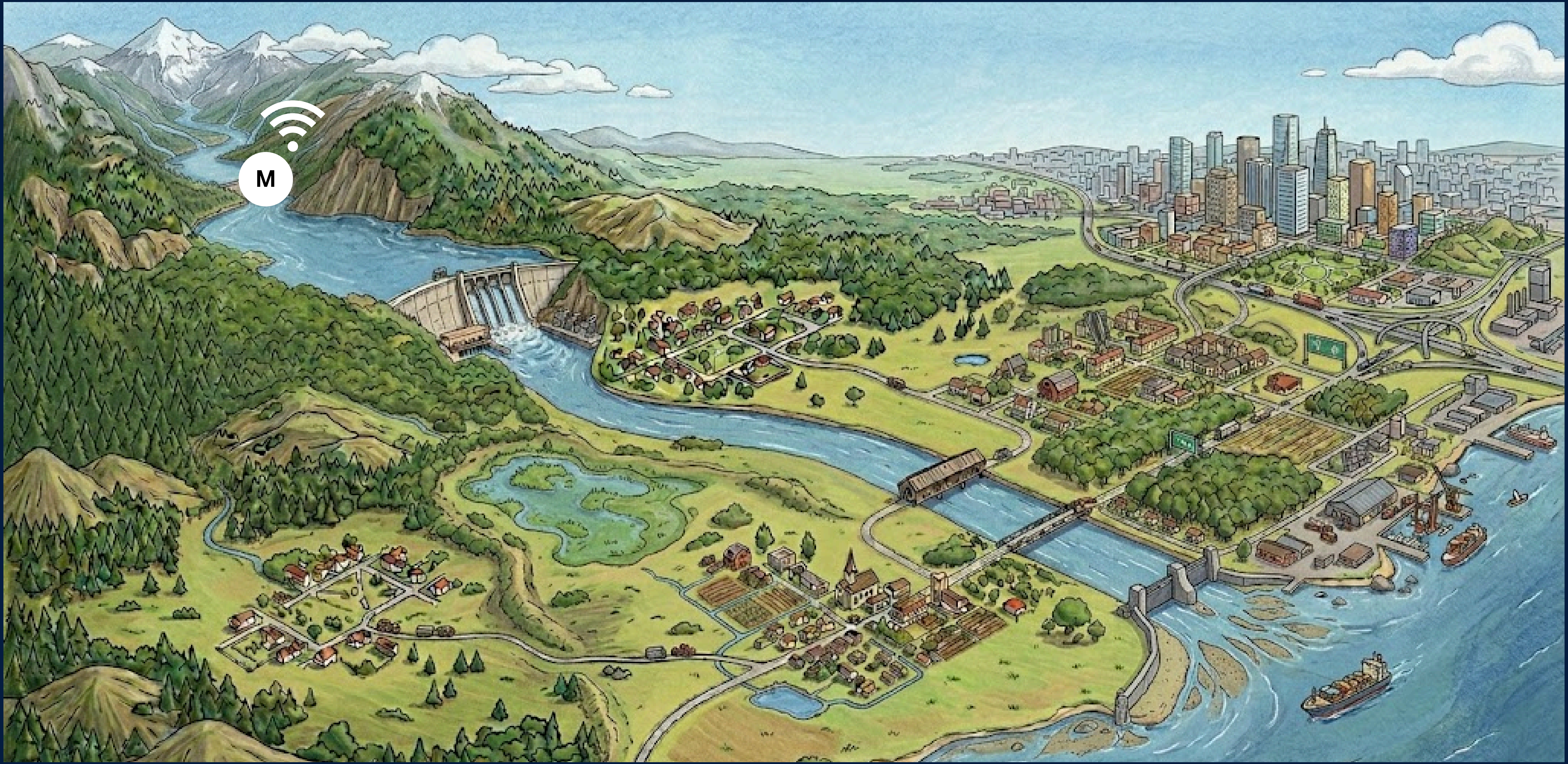
Introduction



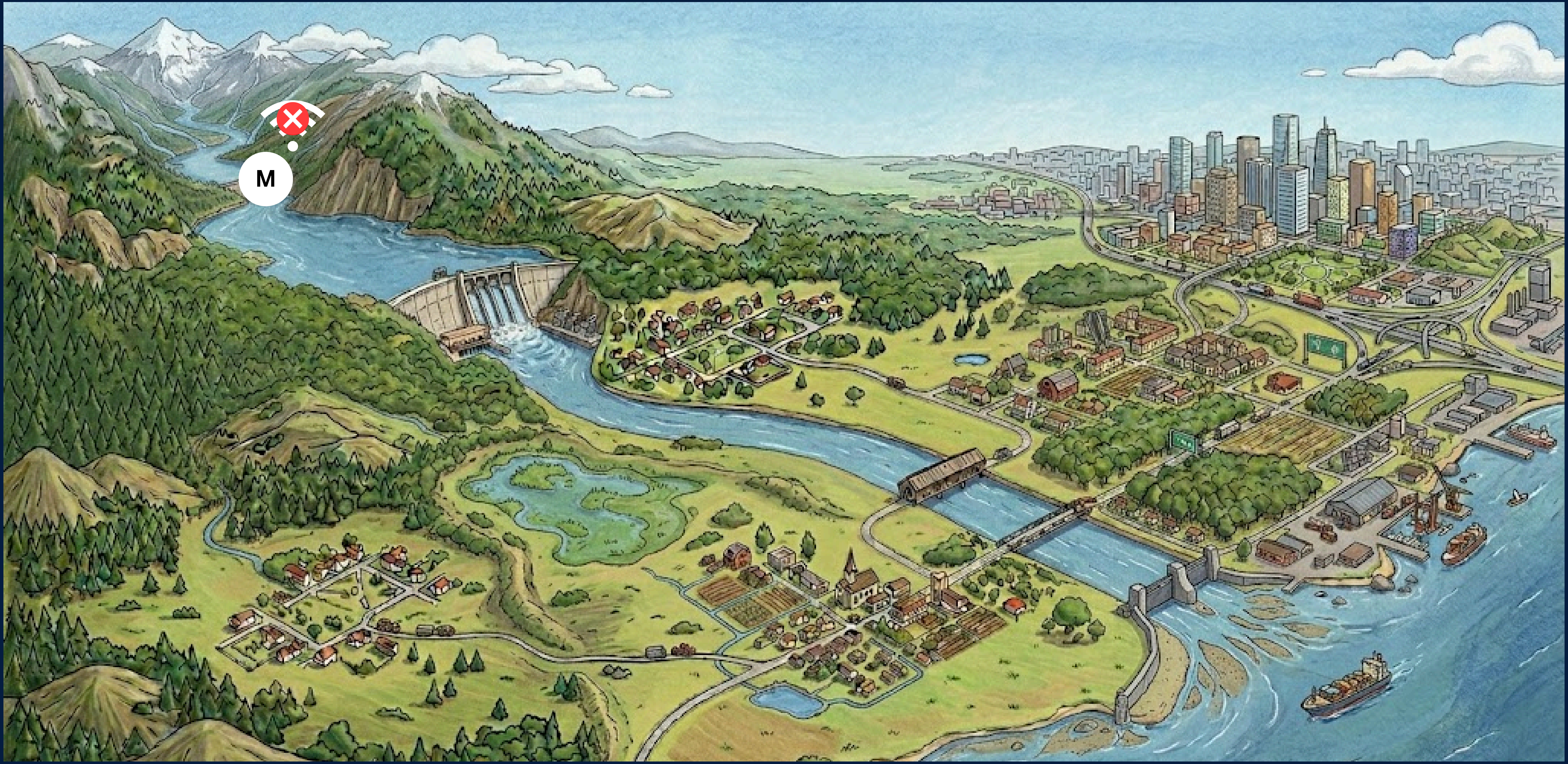
Here, we present an example of an area that can be impacted by flooding.



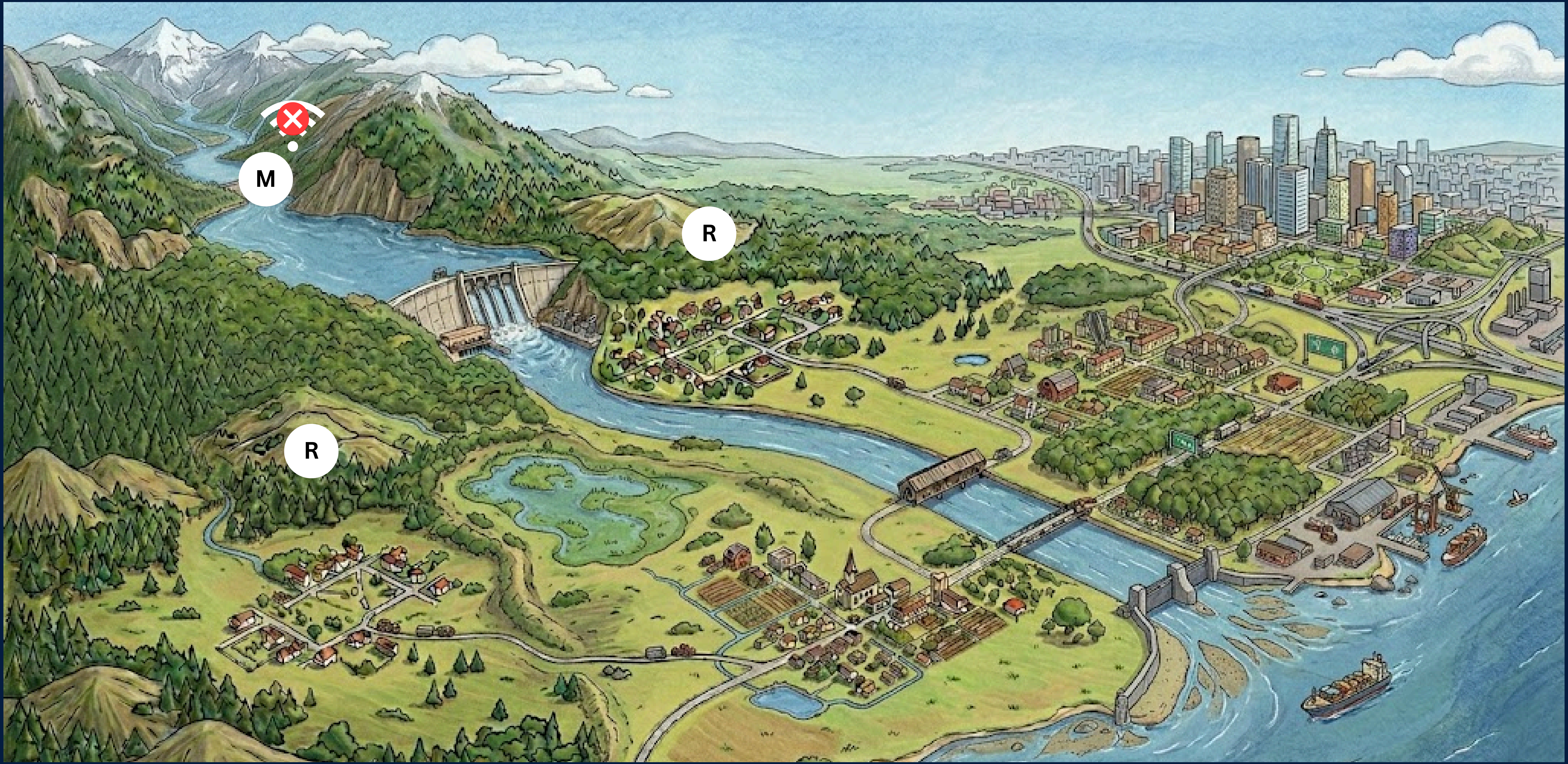
The sensory node is usually deployed near the upstream river of the dam for data collection.



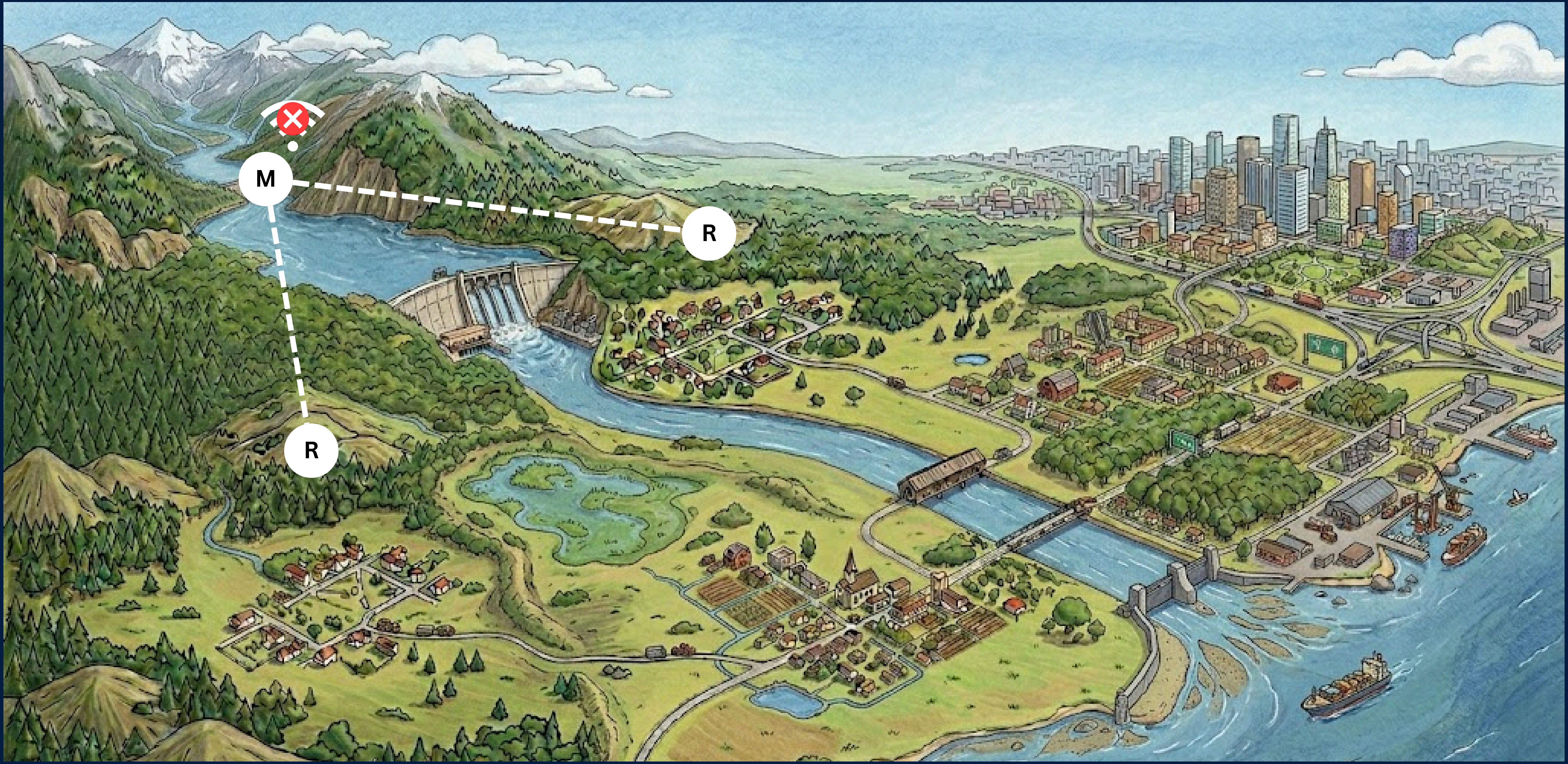
The sensory node can transmit data to the central dashboard using WiFi.



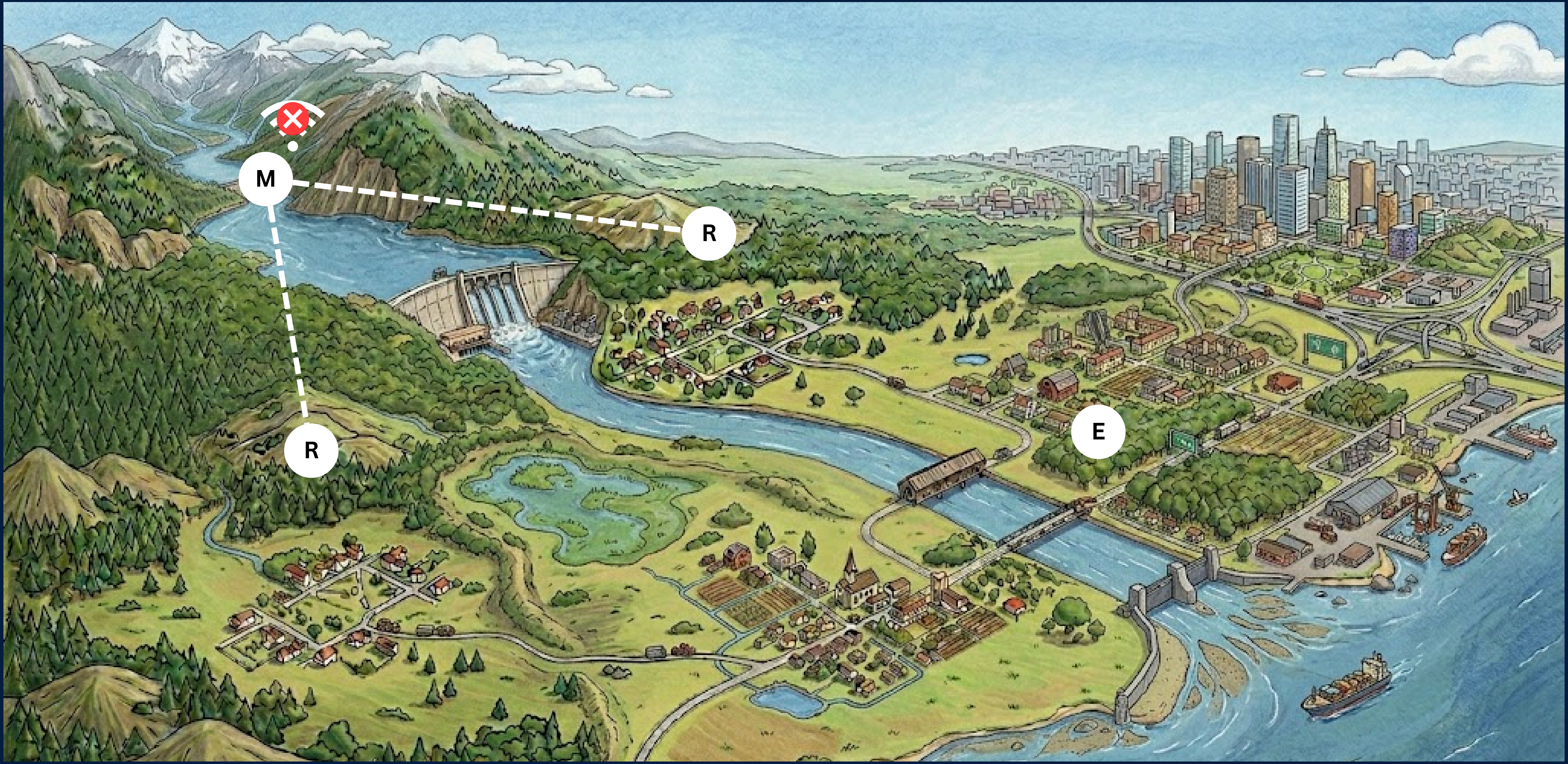
Sometimes, internet access may be interrupted, requiring alternative communication methods.



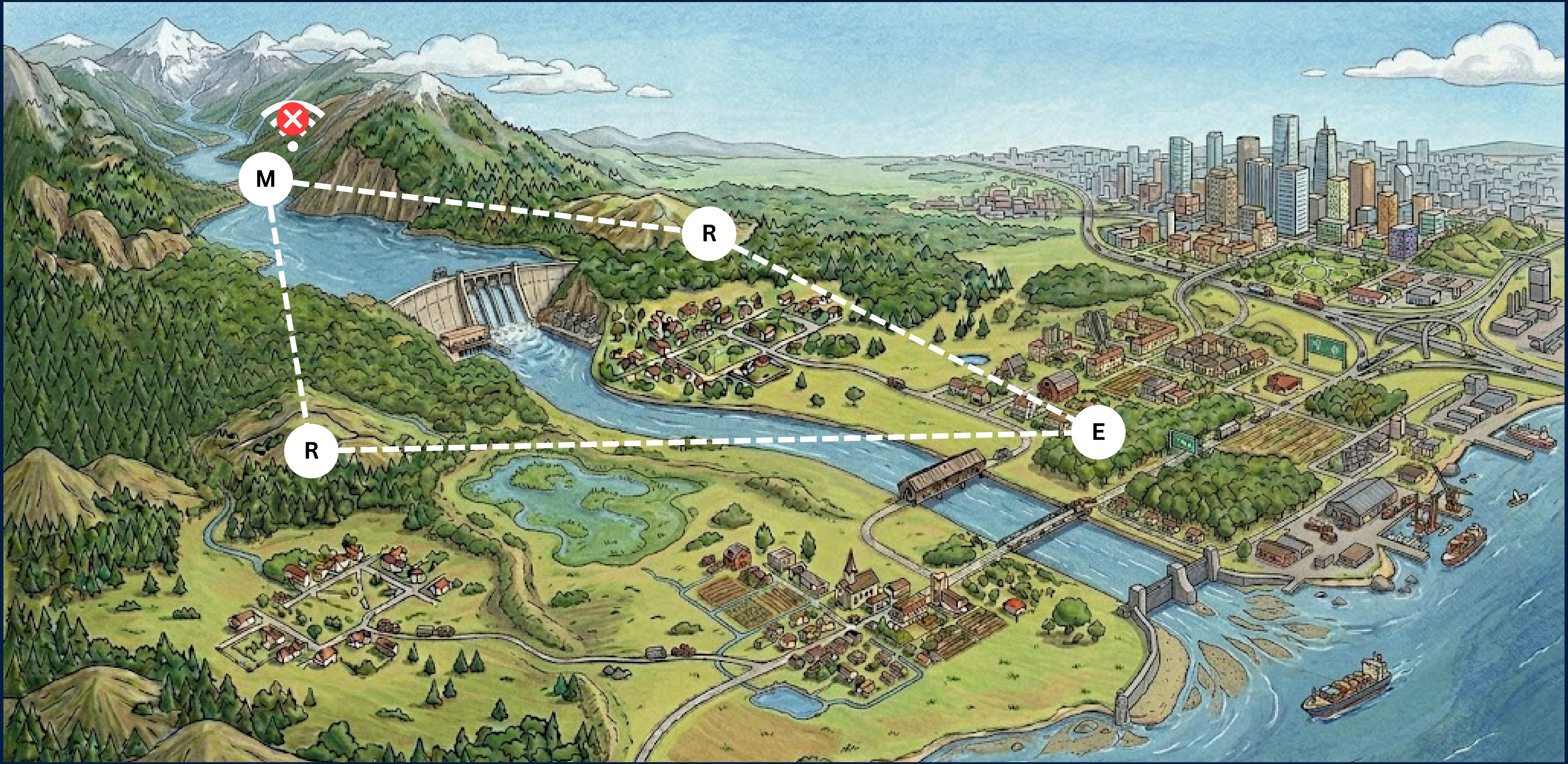
The repeater nodes are placed around the sensory node in a long radius.



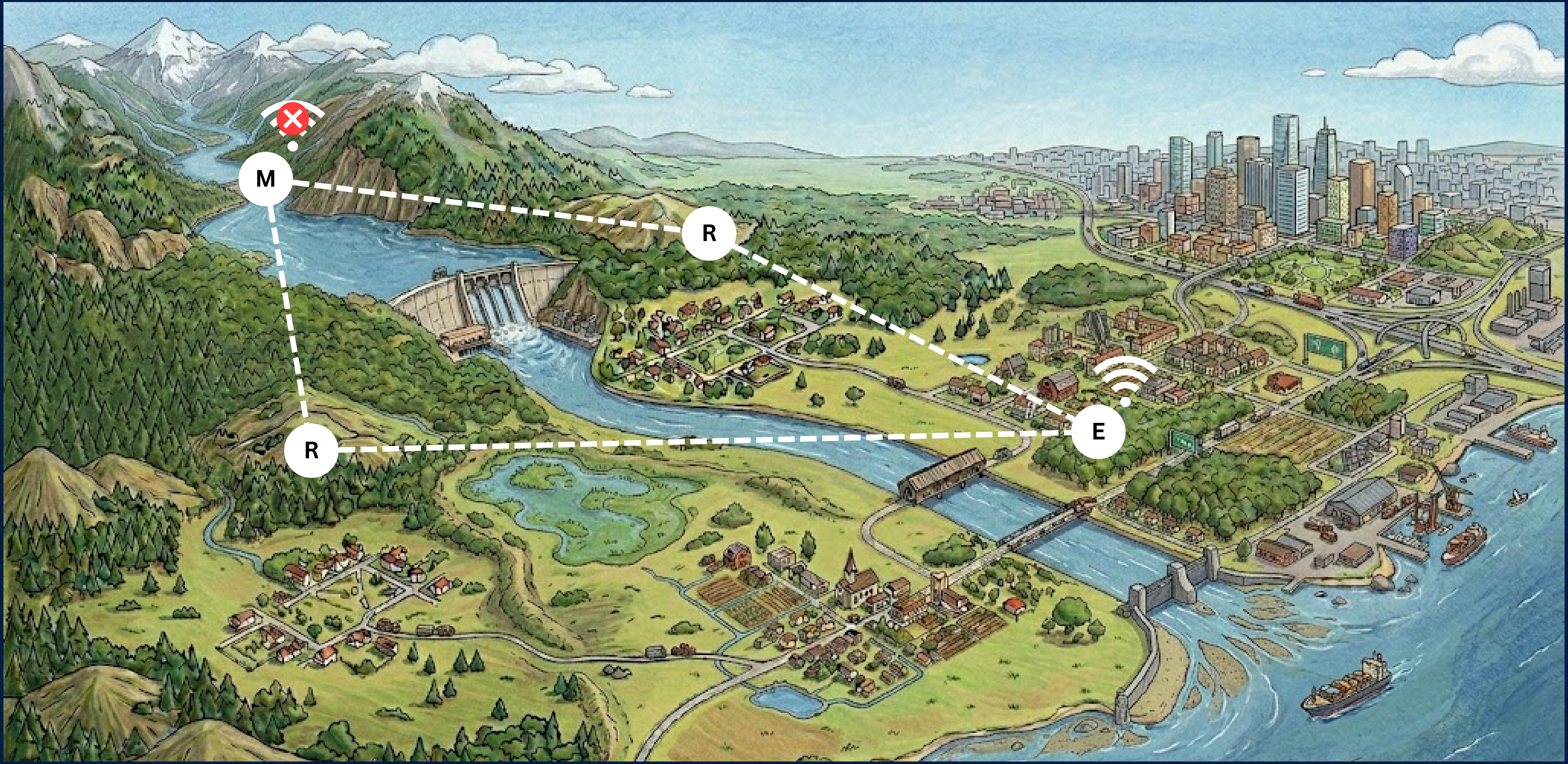
When a sensory node loses internet, it switches to LoRa technology to send data to repeater nodes.



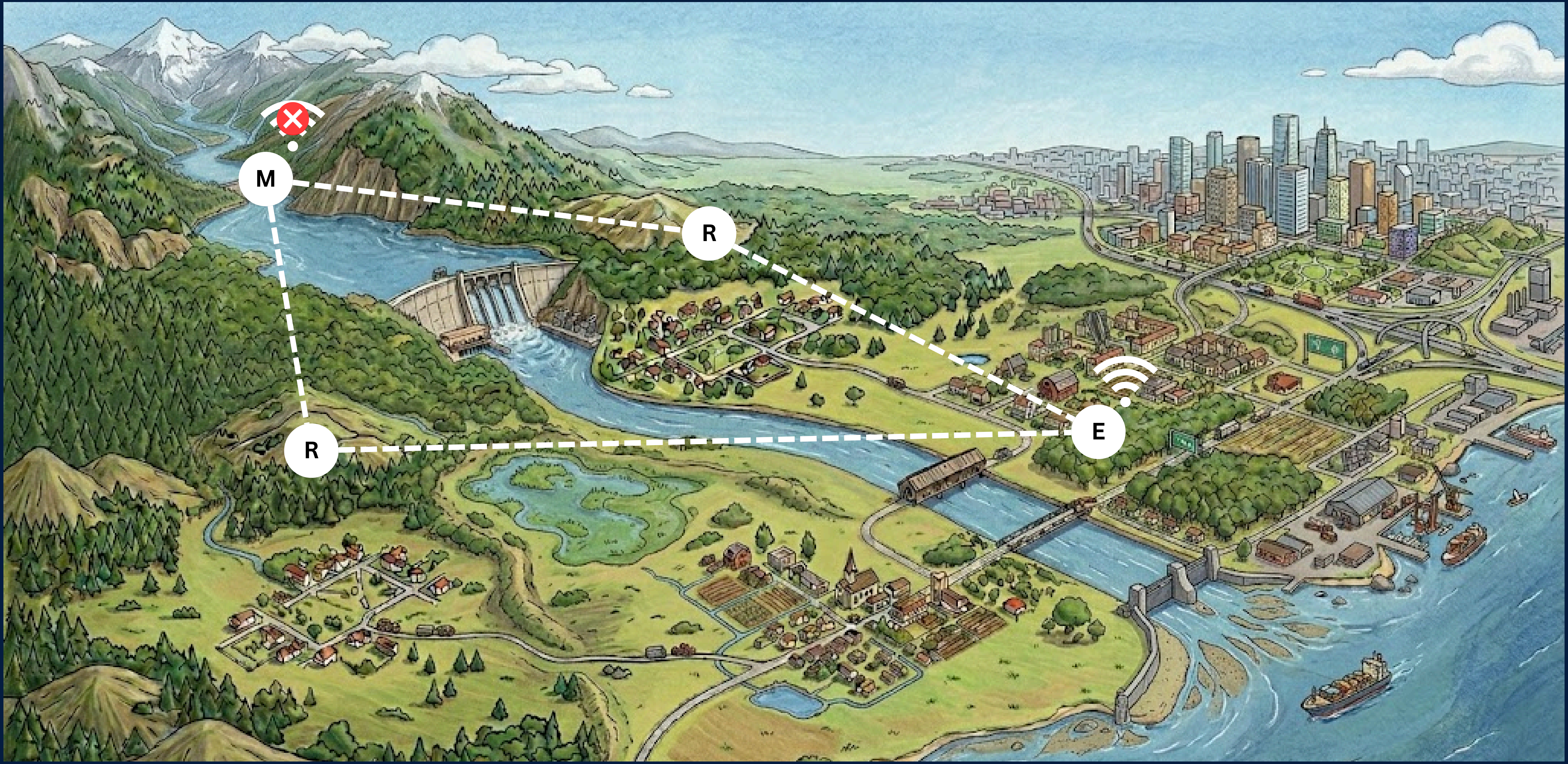
The end nodes are usually positioned in secure areas that are protected from flooding.



The repeater nodes will relay the broadcasts they receive from the sensory node to the end devices.



End devices in safe zones can send data from sensory nodes to the central dashboard via WiFi.



End devices in safe zones can send data from sensory nodes to the central dashboard via WiFi.

Problem Statement & Stakeholders

Context:

- IoT-Based Flood Detection & Disaster Management System with prediction capabilities

Stakeholders:

- Disaster management authorities & Emergency response teams (Primary)
- Flood-affected communities in low-lying areas (Secondary)

Distributed IoT Nodes & power management

MAIN SENSOR NODE

Highest draw

139 mA avg

WiFi always on · GPS · 7 sensors · adaptive TX

29.6h	46.8h	∞
NO SOLAR	LORA FALLBACK	WITH SOLAR

WiFi idle	80mA
NEO-6M GPS	35mA
7 sensors	22mA
WiFi TX burst	2mA avg

WiFi radio = 58% of all power. When WiFi fails during flood, node drops to 88mA — **37% more efficient** automatically.

REPEATER NODE

Most efficient

93 mA avg

Pure LoRa relay · no sensors · no GPS

44.3h	44.3h	∞
NO SOLAR	ALWAYS LORA	WITH SOLAR

ESP32 idle	80mA
LoRa RX listen	11mA
LoRa TX relay	2mA avg
Sensors / GPS	none

49% longer runtime than main node on identical pack. Extends mesh range +2km per hop. Solar surplus fully recharges pack in **21h under load.**

END NODE

Alert gateway

121 mA avg

WiFi bridge · LoRa RX · DFPlayer audio alerts

34.0h	60s	∞
NO SOLAR	ALERT POLL	WITH SOLAR

ESP32 WiFi	80mA
DFPlayer idle	20mA
LoRa RX	11mA
Speaker avg	6mA avg

Dual-path alert delivery — LoRa mesh in, WiFi API out. Audio alert within **60s** of sensor trigger. No single point of failure.

13.3h
SOLAR CHARGE — NO LOAD
1,440mAh ÷ 108mA charge rate

21.4h
SOLAR CHARGE — UNDER LOAD
Net 67mA charge while node running

400mW
USEFUL SOLAR PER NODE
5V 500mA × 80% efficiency · covers full load

1.2W
TOTAL SYSTEM · 3 NODES
1.5W solar total · net positive in sunlight

Distributed IoT Nodes & power management

ADAPTIVE TX INTERVAL — MAIN SENSOR NODE

BATTERY	TX EVERY	AVG CURRENT	RUNTIME PROJECTION	RUNTIME
> 80%	60 s	143.7 mA		28.6h
> 60%	120 s	140.3 mA		29.3h
> 40%	300 s	138.3 mA		29.8h
> 20%	600 s	137.7 mA		29.9h
≤ 20%	1800 s	137.2 mA		30.0h

TX interval increases 30× from full to critical battery (60s → 1800s). Daily TX events drop from 1,440 → 48 at critical tier — 97% fewer transmissions when it matters most.

SAVINGS BREAKDOWN

+2.7h
TOTAL RUNTIME GAINED
vs flat 60s interval throughout. Adaptive TX saves **375 mAh/day**

309 mAh
TX ENERGY SAVED/DAY
1,440 → 48 TX events at critical tier × 5s × 160mA

66 mAh
IDLE ENERGY SAVED/DAY
137mA × 1,740s extra idle time at lowest tier

97%
TX REDUCTION AT ≤20%
System keeps reporting at 30min intervals even at near-dead battery

Key insight: WiFi idle (80mA constant) dominates over TX burst cost — this is why adaptive intervals save only 2.7h rather than the 10h+ you might expect. The real efficiency lever is the automatic LoRa fallback (88mA avg vs 139mA) — a 37% drop that happens automatically when flood conditions take down WiFi infrastructure.

Distributed IoT Nodes & power management

PACKET COMPRESSION — JSON VS BINARY STRUCT

WITHOUT COMPRESSION	WITH BINARY STRUCT
PAYLOAD SIZE ~580 bytes JSON string encoding	PAYLOAD SIZE ~45 bytes Packed C struct
LORA LIMIT EXCEEDS 255B Fragmentation required	LORA LIMIT Fits easily Single packet, no fragmentation
AIRTIME AT SF9 ~4.1s per TX 2 packets minimum	AIRTIME AT SF9 ~0.5s per TX 1 packet, clean delivery
ENERGY PER TX ~5.38 mWh 120mA × 3.3V × 4.1s × 2 pkts	ENERGY PER TX ~0.07 mWh 120mA × 3.3V × 0.5s

92%
SIZE REDUCTION
(580−45)÷580

624 mAh
SAVED PER DAY
1,440 TX × 1.43mWh saved ÷ 3.3V

88%
AIRTIME REDUCTION
Lower collision risk in mesh

DUPLICATE PACKET PREVENTION — 15-ENTRY CACHE

- **Main node** → Sends 1 packet with timestamp (ts) + sequence number — 1 TX
- **Repeater 1** → Receives, checks cache — not seen → forwards once, caches ts+seq — 1 forward
- **Repeater 2** → Receives same packet via 2 paths — second copy dropped by cache — 1 dropped
- **End node** → Receives exactly 1 clean packet → 1 API write to Laravel dashboard — 1 API call

2,880
TX PREVENTED/DAY
4-node mesh · 1,440 pkts/day · 2 dupes each

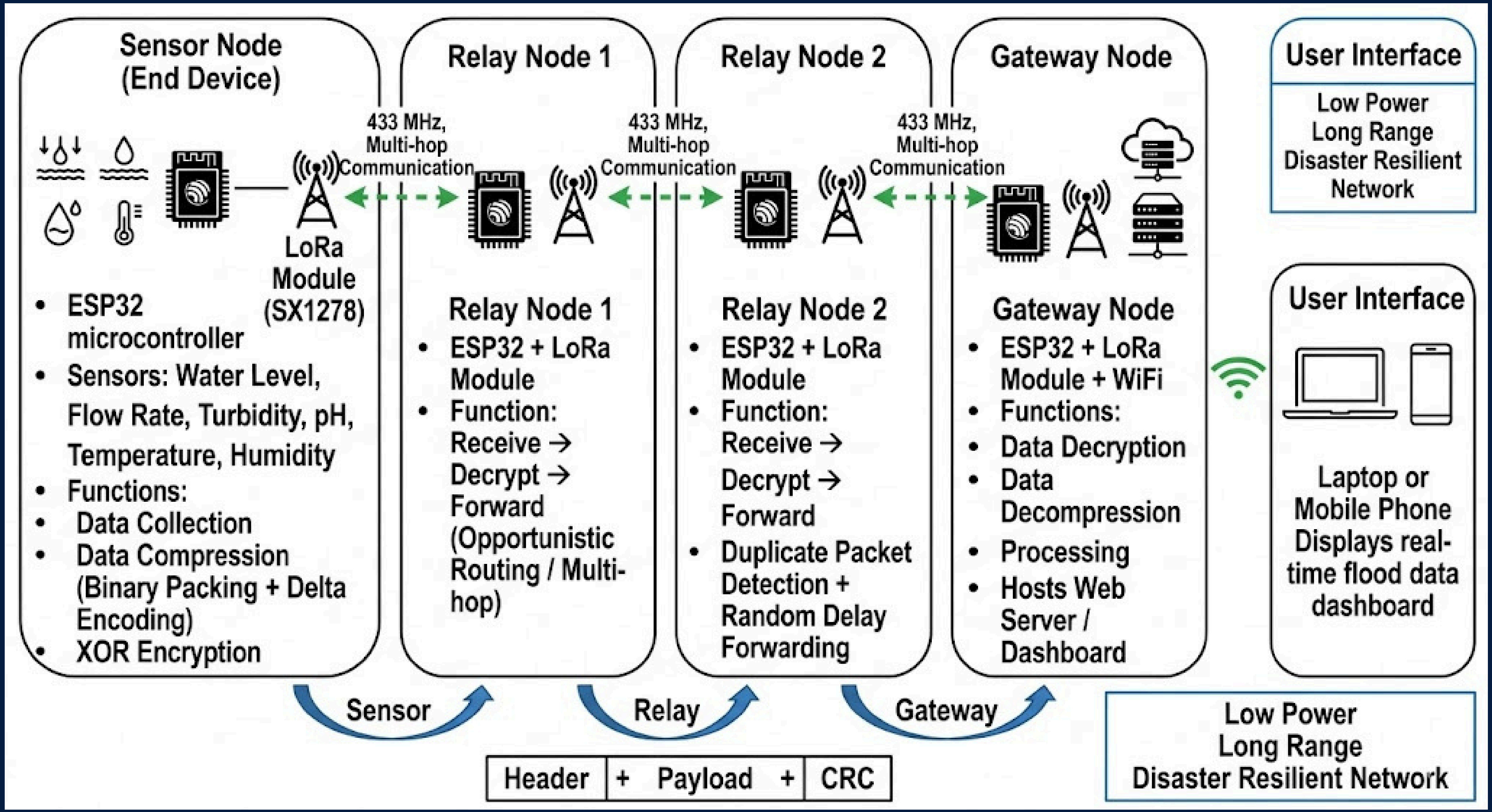
13.3 mAh
SAVED PER NODE/DAY
2,880 × 0.5s × 120mA ÷ 3,600

0
DUPLICATE API WRITES
Dashboard receives clean single-write stream

Without dedup: every sensor packet floods all 3 repeaters up to 3× each — 4,320 LoRa TX events/day instead of 1,440. **Cache reduces mesh radio traffic by 67%** while guaranteeing delivery. Laravel dashboard receives a clean, deduplicated telemetry stream with zero duplicate records.

Component Overview

LoRa Routing, Encryption & Compression

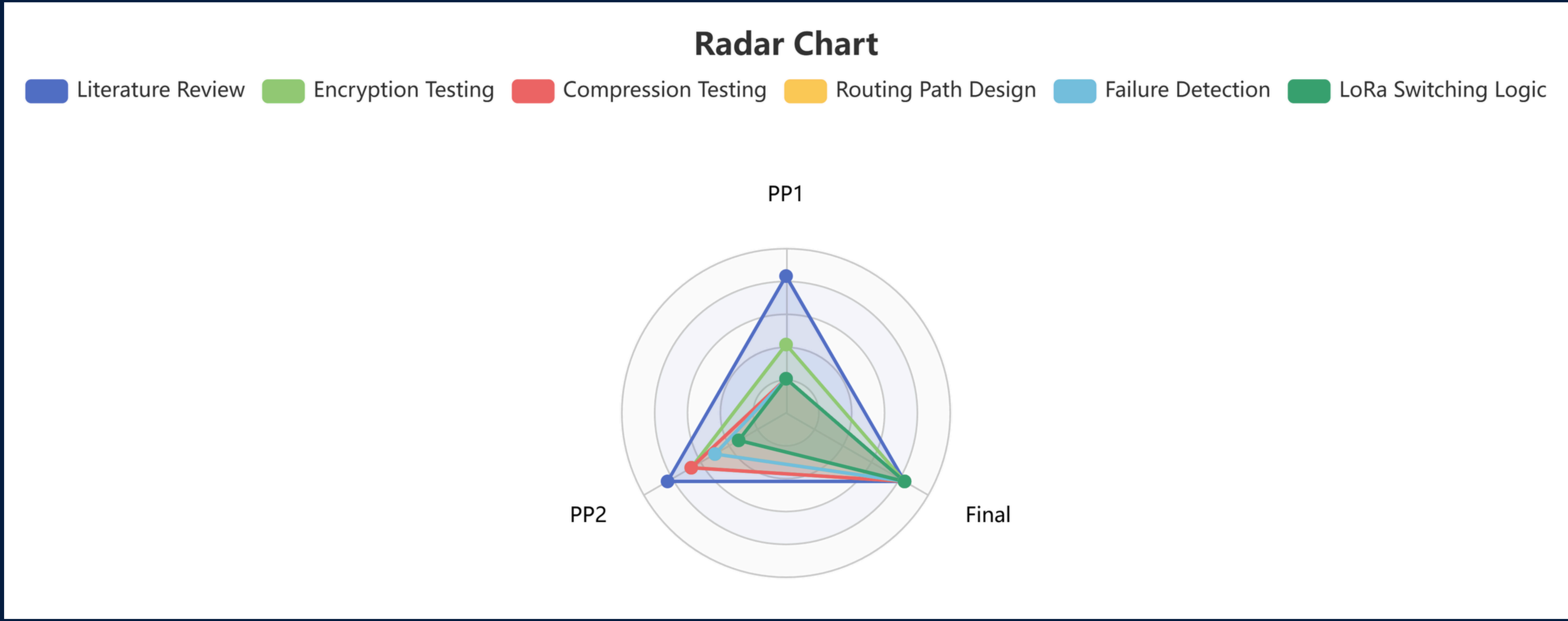


Silva P.K.A
IT22571298

Progress of Research Components: PP1 → PP2 → Final

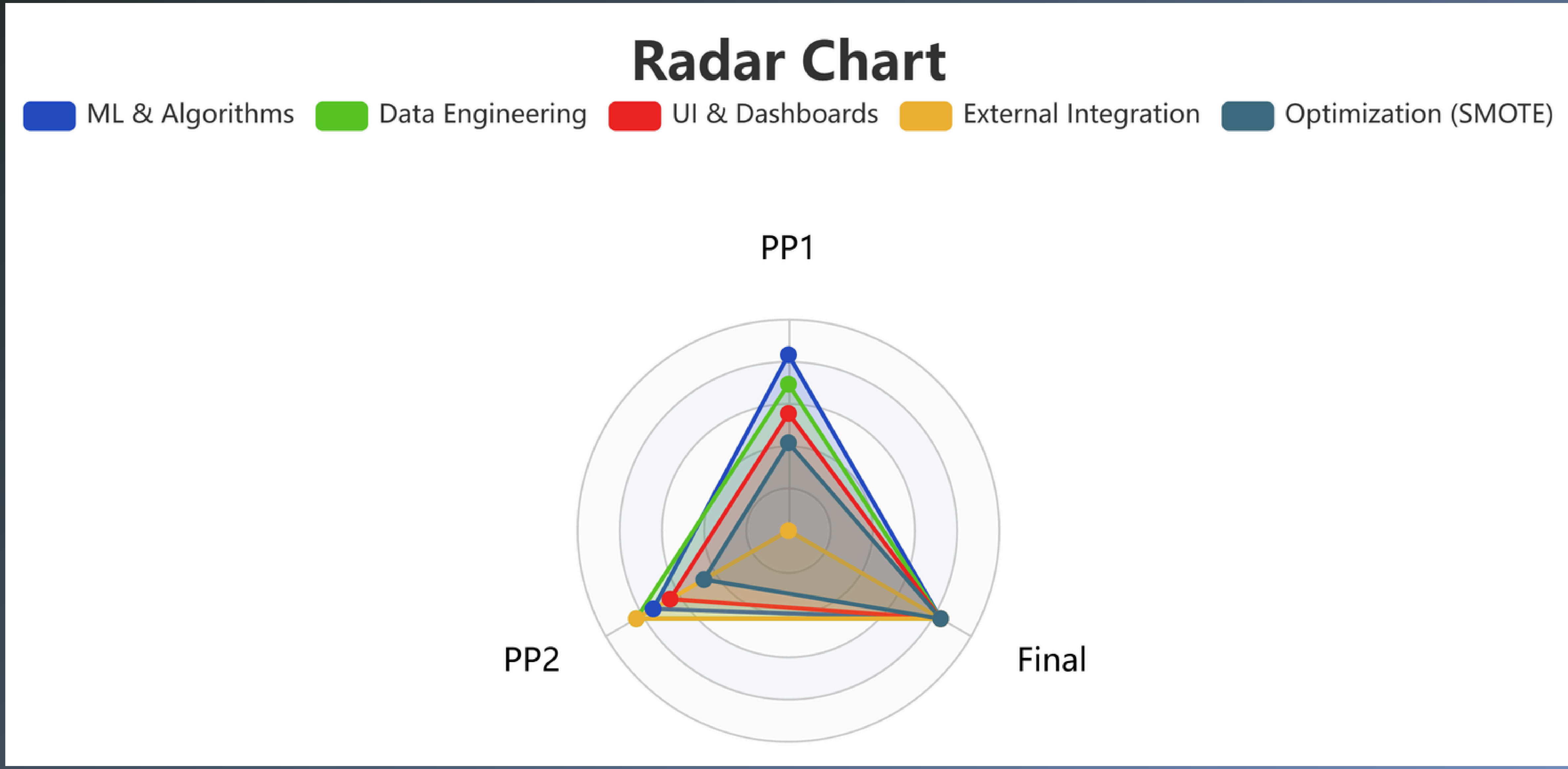
LoRa Routing, Encryption & Compression

Silva P.K.A
IT22571298



Component Overview

Flood Prediction and End-User Dashboard

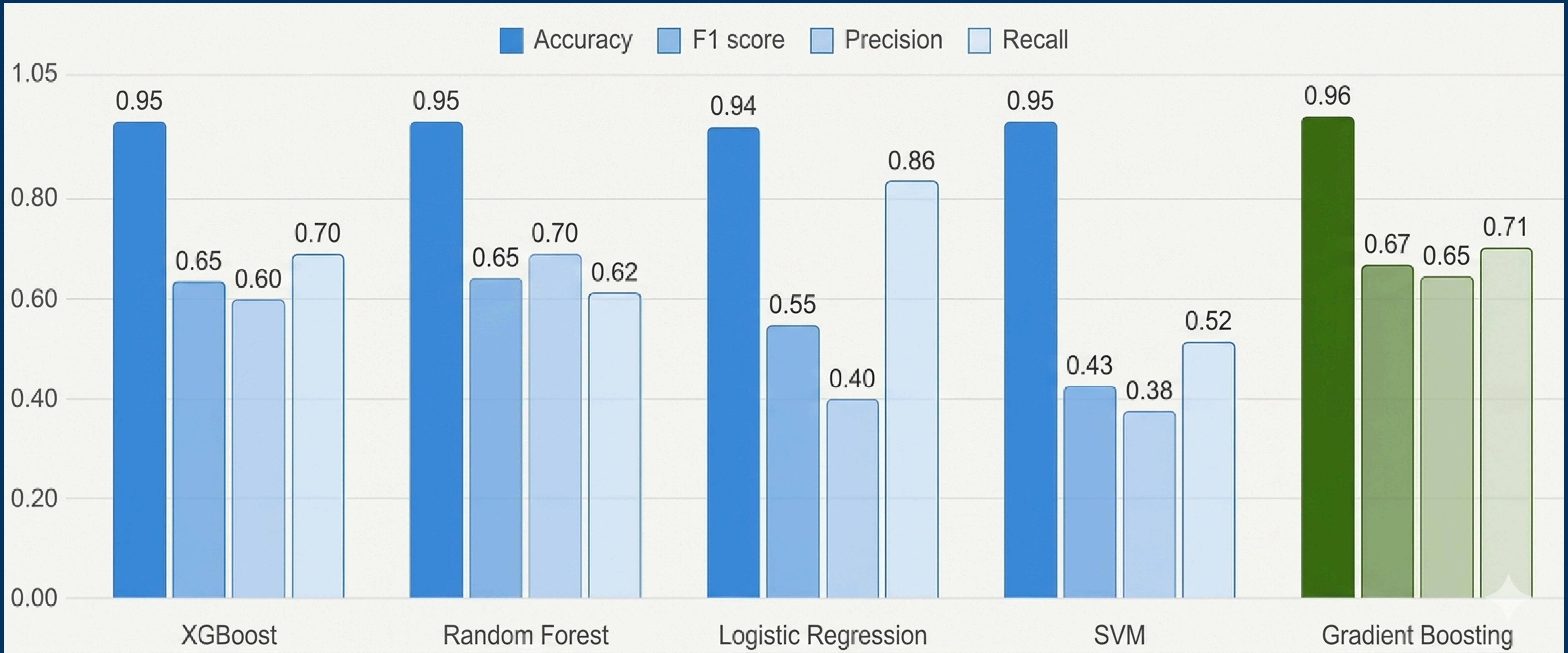


JAYANATH D.T.
IT22563446

Component Overview

Flood Prediction and End-User Dashboard

JAYANATH D.T.
IT22563446



Improvement after PP1

Lakshitha D H
IT22617446

Main dashboard and management system:

- **Real-time Node Monitoring:** - Sensor readings, location, and node health.
- **Sound Alert System:** - Integrated with nodes for local alerts.
- **Log Management:** - Capturing and visualizing node and system logs.
- **Emergency Alert Handling:** - Submission and management of user-reported alerts.
- **SMS Alert Mechanism:** - Sending alerts to predefined groups.
- **User Management:** - Role-based access and control implemented.
- **Rivernet.lk Integration:** - Access to live river data from the main dashboard.

COMMERCIALIZATION / IMPACT

Target Users / Market:

Disaster management authorities, emergency response teams, smart city administrators, and IoT solution providers.

Value Proposition:

Low-cost, IoT-ready flood monitoring and disaster management system with real-time alerts, predictive analytics, and offline network support.

Cost and Feasibility:

Affordable deployment using ESP32-based IoT nodes; scalable system architecture with low maintenance and subscription-based analytics for sustainability.

Adoption Plan:

Pilot projects with local authorities → Integration with smart city platforms → Subscription and licensing for broader rollout.

THANK YOU